

MAE SARIANG, Thailand

WMO: 483250

Lat: 18.175N

Lon: 97.934E

Elev: 213

StdP: 98.79

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	13.2	9.6	7.6	19.2	11.0	8.4	19.7	3.2	29.9	2.5	29.5	0.0	N/A	0.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.8	38.5	24.4	37.4	24.0	36.2	23.8	27.1	32.6	26.7	32.0	26.4	31.6	0.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.6	29.1	25.5	21.2	28.8	25.2	20.8	28.4	86.9	32.8	85.3	32.2	84.0	31.9	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.4	2.8	1.7	DB	9.4	40.5	1.8	1.2	8.1	41.3	7.0	42.1	6.0	42.8	4.7	43.7
			WB	8.7	28.6	1.7	1.0	7.5	29.3	6.5	29.9	5.5	30.4	4.2	31.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	23.2	24.2	27.2	30.3	29.6	27.8	27.1	27.0	27.7	27.8	26.2	23.8
	DBStd	2.62	1.69	1.83	1.75	1.70	2.06	1.38	1.17	1.09	1.09	1.11	1.78	2.12
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6148	410	399	533	608	607	534	529	528	532	551	487	428
	CDD18.3	3107	152	166	275	358	349	284	271	269	282	292	237	171
	CDH23.3	29484	1102	1675	3173	4561	4027	2687	2299	2216	2488	2423	1724	1109
	CDH26.7	12831	439	868	1861	2689	1987	884	621	571	822	942	729	416
(14) Wind	WSAvg	0.1	0.1	0.1	0.2	0.3	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1
(15) Precipitation	PrecAvg	1282	10	2	6	32	190	177	218	261	209	141	20	15
	PrecMax	1606	97	23	69	93	321	330	362	418	378	288	86	83
	PrecMin	1080	0	0	0	0	29	94	140	141	76	36	0	0
	PrecStd	142	25	6	18	32	74	58	62	69	80	78	24	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	36.3	38.2	40.2	39.9	35.0	33.8	33.1	34.2	34.5	34.1	33.0
		MCWB	22.0	21.9	22.6	24.5	25.4	26.4	26.3	26.3	26.6	26.1	25.2	23.8
	2%	DB	32.1	34.9	37.2	39.0	37.9	33.4	32.1	31.8	33.0	33.5	33.1	31.7
		MCWB	21.7	21.7	22.4	24.5	25.6	26.2	25.9	26.0	26.4	26.1	24.9	23.1
	5%	DB	30.9	33.6	36.2	38.0	36.2	32.1	30.9	30.7	32.0	32.6	32.2	30.6
		MCWB	21.5	21.4	22.4	24.4	25.6	25.9	25.7	25.8	26.2	25.9	24.5	22.7
	10%	DB	29.5	32.2	35.1	36.8	34.5	30.8	29.8	29.7	30.9	31.6	31.0	29.3
		MCWB	20.9	21.0	22.2	24.3	25.6	25.7	25.4	25.5	25.9	25.7	24.1	21.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.6	25.4	27.1	27.5	27.2	27.0	27.0	27.4	27.2	26.7	25.2
		MCDB	30.5	32.9	34.8	37.1	34.1	32.1	31.7	31.3	32.2	32.5	32.3	30.9
	2%	WB	23.2	23.3	24.6	26.2	26.8	26.7	26.4	26.5	26.8	26.7	25.9	24.2
		MCDB	29.3	31.8	33.7	35.5	33.5	31.4	30.6	30.4	31.5	31.9	31.2	29.9
	5%	WB	22.4	22.5	23.9	25.7	26.4	26.3	26.0	26.1	26.4	26.3	25.3	23.5
		MCDB	28.4	30.8	32.5	34.5	33.0	30.6	29.8	29.7	30.8	31.3	30.4	28.8
	10%	WB	21.7	21.7	23.2	25.2	26.0	26.0	25.6	25.7	26.0	25.9	24.7	22.7
		MCDB	27.5	29.8	31.6	33.5	32.5	29.9	29.0	28.9	30.2	30.7	29.3	27.6
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	18.9	17.4	13.8	9.4	6.5	5.8	5.8	7.2	9.0	11.4	13.5
		MCDBR	16.9	19.9	18.5	15.3	12.2	8.4	7.6	7.4	8.6	10.2	12.0	13.9
	5% WB	MCWBR	7.1	7.4	5.7	3.8	2.8	2.6	2.6	2.7	3.0	3.7	4.4	5.7
		MCWBR	14.7	17.6	15.6	13.2	9.9	7.2	6.6	6.6	8.0	9.2	10.6	12.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.419	0.480	0.639	0.655	0.523	0.456	0.442	0.449	0.469	0.511	0.437	0.420
		taud	2.259	2.099	1.708	1.742	2.121	2.348	2.390	2.358	2.284	2.120	2.287	2.284
	Ebn at Noon	Ebn at Noon	859	823	709	701	791	838	851	851	835	785	835	845
		Edh at Noon	129	159	243	236	160	126	121	126	135	154	125	123
(44) All-Sky Solar Radiation	RadAvg	RadAvg	4.79	5.59	5.90	6.09	5.15	4.14	3.79	3.81	4.34	4.61	4.76	4.52
	RadStd	RadStd	0.22	0.17	0.37	0.40	0.47	0.36	0.44	0.28	0.20	0.26	0.31	0.24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	+0.32	N/A	N/A	N/A	+0.23	+0.37	N/A	N/A	N/A	N/A
		+0.27	N/A	N/A	N/A	+0.12	+0.19	N/A	N/A	+147	+148

Nomenclature: See separate page